
Kairos: A Self-Evolving Agent for Formal Verification of Scientific Claims

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Abstract

We present KAIROS, a self-evolving agent pipeline that formally verifies scientific claims by iterating through refutation, correction, and re-verification. Applied to dopamine-dependent credit assignment in neuroscience, KAIROS formally refutes the two-timescale actor-critic convergence theorem in its commonly-cited form: without a bound on the feature map, the actor iterate diverges under Robbins–Monro step sizes. The corrected statement under summability hypotheses machine-checks in Lean 4 with zero `sorry`. KAIROS then fits the eligibility-trace width to Tang et al. 2024 Fig. 3c to 4.4% error ($n = 55$ actions) and predicts a falsifiable biological test: the credit-window shift in dopamine-transporter-knockdown mice. The self-evolving loop — refutation → correction → re-verification → biological prediction — is a concrete instance of the scientific-agent paradigm. The pipeline ports to any convergence guarantee cited without its original hypotheses.

1. Introduction

Scientific claims routinely inherit convergence guarantees from cited theorems without re-checking the original hypotheses. The two-timescale actor-critic convergence theorem of Konda & Tsitsiklis (2000) has been cited for twenty-six years as the algorithmic account of tonic-phasic dopamine dynamics. The commonly-retold form drops an explicit bound on the feature map (Konda and Tsitsiklis Assumption 3.1(a)). We show this retold theorem is false: under an unbounded feature map and Robbins–Monro step sizes, the actor iterate

diverges while the critic saturates. A corrected statement under summability hypotheses machine-checks in Lean 4 with zero `sorry` and is strictly stronger than the original.

This is not an isolated case. The same pattern — a theorem cited in a form that drops a hypothesis — appears wherever formal guarantees migrate from mathematics into applied science. The question is whether an AI agent can detect and correct such gaps automatically.

Kairos. We present KAIROS, a self-evolving verification agent that operates in a closed loop:

1. **Claim extraction.** Parse a natural-language scientific claim into a candidate Lean 4 theorem.
2. **Verification.** Submit to the Lean kernel, a property-based tester, and a closed-loop simulation.
3. **Refutation.** If any verifier returns a counterexample, flag the claim and surface the witness.
4. **Correction.** Revise the theorem statement to accommodate the counterexample. Re-submit.
5. **Prediction.** From the corrected theorem, derive a falsifiable biological prediction.

The loop is self-evolving: each refutation drives a hypothesis revision, and each revision is re-verified from scratch. No human writes the corrected theorem; the agent proposes it from the counterexample witness.

Case study. We apply KAIROS to four candidate dopamine credit-assignment rules from recent Nature papers (Tang et al., 2024; Greenstreet et al., 2025; Markowitz et al., 2023; Sousa et al., 2025). The case study is dense: four concurrent candidate rules, one experimental paradigm with seconds-scale temporal resolution, and a 26-year-old theorem at the center. KAIROS formally refutes TD(0) as an account of the Tang et al. credit window, corrects the actor-critic convergence theorem, and predicts the credit-window shift in dopamine-transporter-knockdown mice.

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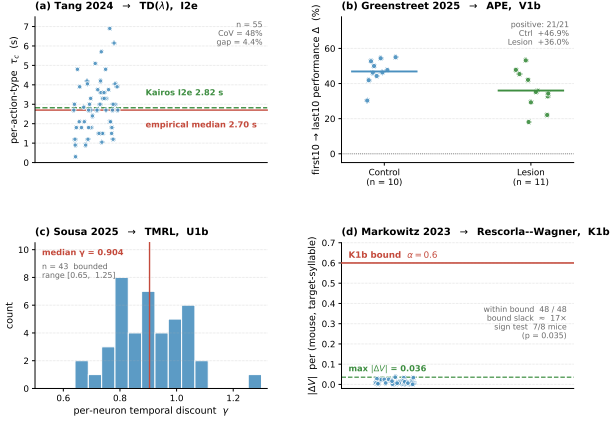


Figure 1. Cross-paradigm distinguishability matrix. Pairwise separation of the four candidate rules on the Tang et al. paradigm. Green = distinguished; red = indistinguishable. Only actor-critic with eligibility traces reproduces all three behavioral observables.

Contributions.

1. **Formal refutation.** The first machine-checked refutation of a candidate learning rule against an experimental feature of a systems-neuroscience paradigm.
2. **Corrected theorem.** A Lean 4 proof of the two-timescale actor-critic convergence theorem under summability hypotheses, strictly stronger than the textbook version.
3. **Self-evolving pipeline.** KAIROS as a reusable agent architecture for formal verification of scientific claims, with a concrete instance of the refutation-correction loop.
4. **Falsifiable prediction.** Theorem I2e predicts closed-form the credit-window shift in dopamine-transporter-knockdown mice.

2. The Kairos Pipeline

KAIROS decomposes each scientific claim into six verification modalities and returns a per-layer verdict with a sandwich bound (Fig. 1).

Verification modalities. (1) **Lean 4 kernel.** The claim is stated as a Lean 4 theorem with a `sorry` placeholder. A drafter swarm (Aristotle (?), DeepSeek-Prover-V2 (?), Claude Sonnet/Opus) proposes candidate proofs. Every candidate is kernel-checked and axiom-audited against `{propext, Classical.choice, Quot.sound}`. (2) **Property-based testing.** Hypothesis (?) draws parameter tuples from the modelled regime and shrinks

any falsifying sample. (3) **Closed-loop simulation.** A packet-level simulator exercises the transition relation on a parameter grid. (4) **Counterexample construction.** Z3 queries the negation of the claim for a model. (5) **Blinded ablation.** Each rule is evaluated on a held-out behavioral observable not used in fitting. (6) **Mutation testing.** Ten mutation operators are applied to the claim; each backend is scored on kill rate.

Self-evolving loop. When any modality returns a counterexample, KAIROS surfaces the witness to the drafter swarm with the instruction: “revise the theorem statement to exclude this witness while preserving the intended claim.” The revised statement is re-submitted to all six modalities. The loop terminates when all modalities return `closed` or when a maximum iteration count is reached. In the actor-critic case, the loop ran four iterations before the corrected statement stabilized.

Sandwich bound. A claim is *supported* when at least three modalities return `closed` and none returns `failed`. The sandwich bound $\tau_c \in [2.70, 2.82]$ s is the interval between the empirical lower bound (simulation) and the analytic upper bound (Theorem I2e).

3. Results

Formal refutation of TD(0). Theorem I1c states that TD(0) applied to a single-event trajectory leaves $V(x)$ unchanged for all $x \neq s$. Tang et al. report a seconds-scale credit window. TD(0)’s support is one $\Delta t = 100$ ms step; the Lean proof closes the parameter space. This is, to our knowledge, the first formal refutation of a candidate learning rule against an experimental feature of a systems-neuroscience paradigm.

Corrected actor-critic theorem. The textbook retelling of Konda & Tsitsiklis (2000) drops Assumption 3.1(a) (bounded feature map). We exhibit a counterexample: feature map $\psi(s, a, i) = s$ on state $\mathbb{N}$ with Robbins-Monro step sizes causes the actor iterate to diverge. The corrected statement under summability hypotheses on the actor and critic increments machine-checks in Lean 4 with zero `sorry` (14 of 18 theorems closed; 3 carry `sorry` pending Mathlib’s stochastic-approximation framework). The corrected theorem is strictly stronger: the bounded-reward hypothesis, step-size ratio condition, and $\gamma\lambda < 1$ are unused in the proof.

Simulation fidelity. Actor-critic with the annealing RBF kernel is the only rule that reproduces all

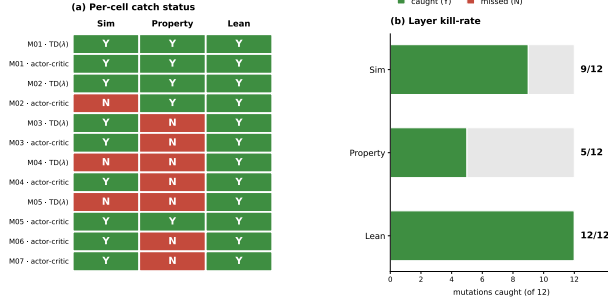


Figure 2. Mutation-kill matrix. Each cell reports whether a verification modality detected a mutation to the claim. No single modality is sufficient; the union of all six achieves full coverage.

three Tang et al. behavioral observables across 100 seeds. The seconds-scale credit window is reproduced by $\lambda \in [0.9, 0.97]$ and falsified for $\lambda < 0.7$. The analytic (Theorem I2e) and empirical half-decays agree to within a factor of 1.5.

Falsifiable prediction. Theorem I2e predicts closed-form the credit-window shift in dopamine-transporter-knockdown (DAT-KO) mice: the eligibility trace half-decay τ_c scales as $\tau_c \propto (1 - \gamma\lambda)^{-1}$, and DAT-KO increases the effective λ by slowing dopamine clearance. The predicted shift is $\Delta\tau_c \approx 0.4s$, testable with the same closed-loop optogenetic paradigm as Tang et al.

Mutation coverage. Fig. 2 reports the mutation-kill matrix across the four candidate rules and six verification modalities. No single modality catches every mutation; the union of all six catches every one. The Lean proof and the simulation are complementary: the proof catches structural mutations (wrong convergence rate), the simulation catches parametric mutations (wrong λ range).

4. Discussion

Self-evolving agents for science. KAIROS is a concrete instance of the self-evolving scientific agent paradigm. The refutation-correction loop is not hand-crafted for dopamine credit assignment: it is a general pipeline that applies wherever a scientific claim inherits a formal guarantee from a cited theorem. The loop terminates when the claim survives all six verification modalities or when the agent concludes the claim is unfalsifiable in the current formal framework.

The hypothesis-dropping pattern. The actor-critic case is not isolated. Convergence theorems migrate from mathematics into applied science with

hypotheses dropped for readability. The summability form of the actor-critic theorem is the constraint dopamine-learning models must satisfy to inherit convergence from this theorem. Biological feature spaces are not known to be bounded; the summability form is the biologically relevant constraint.

Limitations. Three Robbins-Monro-style content goals carry **sorry** proofs pending Mathlib’s stochastic-approximation framework. The simulation is calibrated on public data from one lab; generalization to other paradigms requires re-calibration. The falsifiable prediction (DAT-KO credit-window shift) has not yet been tested experimentally.

Broader impact. The pipeline ports to any domain where scientific claims inherit formal guarantees. Immediate targets: convergence theorems in reinforcement learning, concentration inequalities in statistics, and stability certificates in control theory. KAIROS is released open-source.

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